

**Certificate of Advanced Studies in Natural Language  
Processing (NLP)**

Final Project Report

**A Fine-Tuned German BERT-base Model to  
Detect Governments' Blame Avoidance and  
Credit Claiming Strategies**

Submitted by:

Céline Honegger

celine.j.honegger@gmail.com

University of Bern

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## Abstract

This final project report for the CAS Natural Language Processing at the University of Bern documents the development and evaluation of a supervised natural language processing model for identifying credit claiming and blame avoidance strategies in governments' COVID-19 media-conferences. The project starts from human-coded segments from Swiss and German governmental communication and fine-tunes deepset/gbert-base, a German BERT-base model, on German-language segments. The central methodological challenge is that the original annotation scheme contains five rhetorically meaningful but partly overlapping labels. The first model therefore uses the five original labels; a second and theoretically more robust model merges them into three broader categories: credit claiming, implicit blame avoidance, and explicit blame shifting.

The five-label model reaches an accuracy of 0.579 and a macro F1 score of 0.499 on the German test set. Inspection of the confusion matrix shows that the model has particular difficulty distinguishing fine-grained variants of responsibility communication, especially shifting responsibility versus explaining or sharing responsibility. After merging the labels, performance improves to an accuracy of 0.609 and a macro F1 score of 0.604. The improvement supports the theoretical decision to model broader blame-avoidance categories rather than forcing the classifier to separate coding distinctions that are substantively close and scarce in the training data.

Finally, the report evaluates whether the German model can be applied to French-language Swiss segments after machine translation into German. The translated-French evaluation obtains an accuracy of 0.649 and a macro F1 score of 0.648. This result suggests that a translation-based pipeline is feasible for this corpus, although it should be interpreted as performance on machine-translated German text rather than direct French-language understanding.

# 1 Introduction

In moments of crisis, governments face heightened public scrutiny. Decisions are made under uncertainty, policy trade-offs are visible, and unpopular restrictions may quickly become politically costly. Under these conditions, political actors often use presentational strategies to protect legitimacy, distribute responsibility, and emphasize successful action (Honegger, 2025; Hood, 2011). Two communication strategies are especially relevant: blame avoidance and credit claiming. Blame avoidance refers to attempts to reduce responsibility for negative or potentially blameworthy outcomes, whereas credit claiming highlights successful decisions, preparedness, or effective implementation. The COVID-19 pandemic offers a particularly important setting for studying these strategies. Governments had to justify restrictions, explain uncertainty, allocate responsibility across federal levels, communicate scientific advice, and claim credit for protective measures or vaccination progress. Media conferences are well suited for this analysis because they combine strategic government messaging with responses to immediate public and journalistic pressure. However, manually coding such texts is time consuming. This limits the scale at which researchers can study credit and blame dynamics across countries, languages, and phases of a crisis.

The guiding question for my final project is thus: how well can a fine-tuned German BERT-base model classify short government communication segments into theoretically meaningful credit-claiming and blame-avoidance categories? A secondary question is whether the trained German model can be extended to French segments through machine translation. These questions are intentionally practical. The goal is not to replace qualitative research, but to assess whether a model can support larger-scale screening and measurement once a high-quality annotated corpus exists.

From an NLP perspective, this task is demanding because the relevant categories are not simply defined by single keywords. A segment can avoid blame without using an overtly negative expression, for example by emphasizing federalism, shared competence, or the responsibilities of cantons and citizens. Similarly, credit claiming may appear through positive self-presentation, references to preparedness, or claims that government action prevented worse outcomes. The model therefore needs to learn rhetorical and contextual patterns rather than only surface vocabulary.

The aim of this project is therefore practical and methodological: to fine-tune a transformer-based language model that can detect governments’ credit claiming and blame avoidance strategies in short text segments. I use deepset/gbert-base as the base model because the primary training corpus is German. The project proceeds in four steps. First, the annotated Excel data are converted into reproducible CSV files. Second, French segments are excluded for the German training run. Third, a five-label GBERT classifier is fine-tuned and evaluated. Fourth, the label scheme is merged into broader theoretical categories and the

model is re-trained. As an additional test, French Swiss segments are translated into German using Helsinki-NLP/opus-mt-fr-de and classified with the fine-tuned merged-label GBERT model.

The report follows the structure of the project workflow. The method section describes data preparation, label construction, model fine-tuning, and evaluation. The data section summarizes the corpus and the distribution of labels and targets. The results section compares the five-label and merged-label classifiers and then evaluates the translated-French segments. The conclusion discusses what the results imply for future large-scale analyses of blame avoidance and credit claiming strategies.

## **2 Method: Fine-Tuning a German BERT-base Model**

### **2.1 Data preparation**

The raw data consist of two Excel workbooks containing human-coded media-conference segments: one for Swiss COVID-19 media conferences and one for German COVID-19 media conferences. This data is then stored in reproducible CSV files where each observation contains a segment identifier, train/test split, country, conference date, target, label, text, and detected language.

Because the base classifier is German, I separate German and French text. Since all German text segments are in German, only the Swiss data is classified with a transparent rule-based language filter that compares German and French marker words. This produces 670 German segments and 222 French segments. The French segments are not used for model fine-tuning, but they are kept for the translation-based external evaluation.

The raw Excel workbooks remain unchanged in the project data folder, and all derived datasets are to be found in the processed-data folder. The script removes empty labels and empty text segments but does not impute missing information or rewrite the annotated categories. Dates, countries, targets, and train/test splits are retained as metadata so that later analyses can be traced back to the source corpus.

### **2.2 Annotation scheme and label merging**

The original coding scheme contains credit claiming and four blame-related labels. For labels with a target, the original code uses a 'target > label' structure. For example, 'Citizens > shifting responsibility' identifies citizens as the target and shifting responsibility as the strategy. The first model uses the five original labels: credit claiming, praising, explaining/sharing responsibility, shifting responsibility, and blaming. After inspecting the first model, I merged conceptually related and empirically difficult-to-separate categories. Praising and explaining/sharing

**Table 1:** Datasets used for model training and evaluation

| Dataset                      | Rows | Training rows | Evaluation rows | Purpose                                                                        |
|------------------------------|------|---------------|-----------------|--------------------------------------------------------------------------------|
| German segments              | 670  | 537           | 133             | Five-label German model                                                        |
| German merged-label segments | 670  | 537           | 133             | Main three-class German model                                                  |
| French segments              | 222  | 0             | 0               | Source rows for translation; not used directly for model fitting or evaluation |
| Translated German segments   | 222  | 0             | 222             | External translated-French evaluation set                                      |

responsibility are combined as implicit blame avoidance. Blaming and shifting responsibility are combined as explicit blame shifting. Credit claiming remains unchanged. This three-class scheme is theoretically more parsimonious and empirically more balanced.

This merging decision is also grounded in the substantive meaning of the strategies. Praising and explaining or sharing responsibility both allow the speaker to avoid direct blame while keeping the tone cooperative or justificatory. They are therefore grouped as implicit blame avoidance. By contrast, blaming and shifting responsibility are more direct ways of locating responsibility outside the central government. They are therefore grouped as explicit blame shifting. The new scheme reduces noise without collapsing the central contrast between claiming credit and avoiding blame.

### 2.3 Model and training setup

The classifier is based on deepset/gbert-base, a German BERT-base model available through the Hugging Face Transformers library. For classification, the model is loaded with a sequence classification head. The input text is tokenized with the corresponding BERT tokenizer, truncated or padded to a maximum sequence length of 256 tokens. The model is fine-tuned for five epochs with a batch size of 8 and a learning rate of  $2e-5$ . The existing train/test split is preserved throughout the analysis, allowing comparable evaluation across label schemes.

**Table 2:** Original labels and merged label categories

| Original label                    | Merged label             | Interpretation                                                                                |
|-----------------------------------|--------------------------|-----------------------------------------------------------------------------------------------|
| credit claiming                   | credit claiming          | The government presents itself as responsible for successful action or positive developments. |
| praising                          | implicit blame avoidance | Responsibility is softened by highlighting others' contributions or positive conduct.         |
| explaining/sharing responsibility | implicit blame avoidance | Responsibility is distributed or contextualized without direct accusation.                    |
| shifting responsibility           | explicit blame shifting  | Responsibility is moved away from the government toward another actor.                        |
| blaming                           | explicit blame shifting  | Another actor is explicitly associated with blameworthy conduct or outcomes.                  |

The training script stores the fine-tuned model, tokenizer, test predictions, confusion matrix, and metrics. Macro F1 is used alongside accuracy because the label distribution is imbalanced. Macro F1 averages performance over classes and is therefore more sensitive to poor performance on smaller categories. The best merged-label checkpoint is selected by macro F1.

No extensive hyperparameter search is conducted. This is appropriate because the main question is whether the task is learnable at all and whether the merged label scheme improves performance. The chosen settings are standard for small BERT fine-tuning tasks: a low learning rate, small batch size, and a modest number of epochs. The same settings are used for both the five-label and merged-label models, which keeps the comparison between label schemes as direct as possible.

## 2.4 Translation-based evaluation

The final step tests whether the German model can be applied to French-language Swiss segments after translation. The 222 French segments are translated into German using Helsinki-NLP/opus-mt-fr-de. The translated text is saved in a separate CSV file and evaluated with the fine-tuned merged-label GBERT model. This procedure does not test direct French-language understanding. Instead, it tests a practical pipeline in which French material is first translated into German and then classified with the German model.

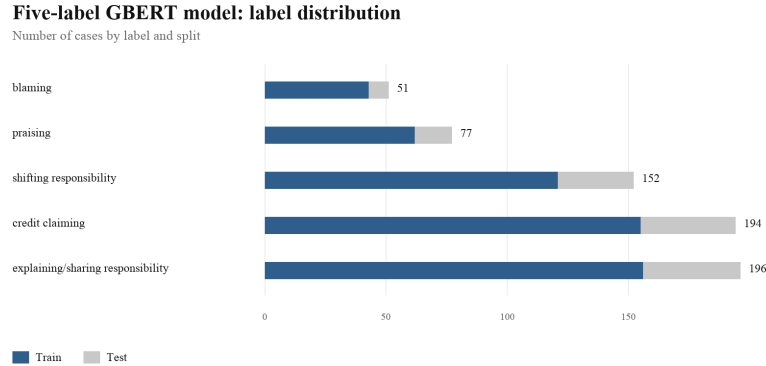
This translation strategy has two advantages. First, it avoids training a second classifier on a much smaller French subset. Second, it uses the same classification head and label definitions as the German experiment, making results easier to compare. The trade-off is that translation may alter the rhetorical form

of the original utterance. For this reason, the translated-French evaluation is treated as a robustness and feasibility test rather than as a definitive measure of French-language model performance.

### 3 Data

The final German training and evaluation corpus contains 670 segments. The five-label version is imbalanced: explaining/sharing responsibility and credit claiming are the largest classes, whereas blaming and praising are substantially smaller. This imbalance is visible in Figure 1 and is important for interpreting the first model’s lower macro F1 score.

The dataset is composed of short, manually selected coded segments rather than full speeches or press-conference transcripts. This is an advantage for supervised classification because each row contains a relatively focused rhetorical move. It is also a limitation because segment boundaries may remove surrounding context that human coders used implicitly. Some ambiguous responsibility statements may therefore be harder for the model than for a human reader who remembers the broader question-and-answer sequence.



*Figure 1. Distribution of the original five labels in the German corpus.*

The merged-label scheme reduces this imbalance. The largest class, implicit blame avoidance, contains 273 observations; explicit blame shifting contains 203 observations; and credit claiming contains 194 observations. The merged scheme therefore retains substantive distinctions while creating a more stable learning problem for the classifier.



### Merged-label GBERT model: label distribution

Number of cases by label and split

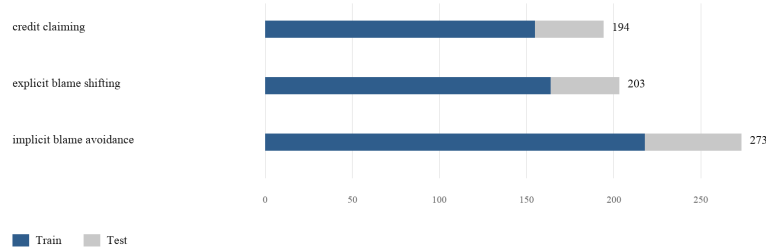


Figure 2. Distribution of merged labels in the German corpus.

The target distribution shows who is named as the responsible actor. In the German corpus, subnational actors are the most frequent targets, followed by citizens, and experts. No target corresponds to the credit claiming strategy. This distribution reflects the institutional context of consensus democracies, where federal and subnational authorities frequently share or negotiate responsibility.

The French subset is smaller but analytically important. It contains 222 segments, all from the Swiss corpus, and includes the same substantive label categories. The translated German version of these rows provides a first test of whether the model can travel across the multilingual Swiss setting. Because French segments are not used for fine-tuning, this evaluation also gives a useful indication of how the classifier behaves beyond the language of its training data.

### Merged-label GBERT model: target distribution

Number of cases by coded target

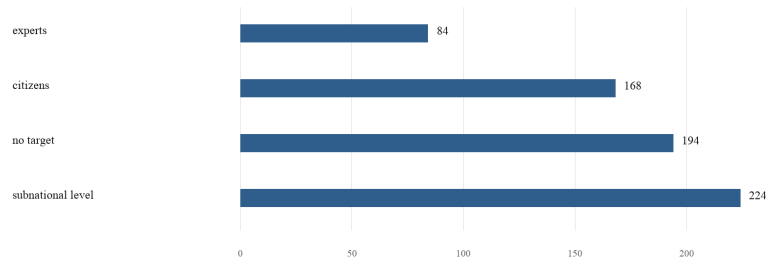


Figure 3. Target distribution in the merged German corpus.

## 4 Results

### 4.1 Five-label baseline

The five-label GBERT model provides a useful baseline. It reaches an accuracy of 0.579 and a macro F1 score of 0.499 on the German test set. The training history shows that the model learns from the data, but the class-level metrics reveal that the fine-grained labels are not equally recoverable from the available training examples. Credit claiming is the strongest class, with an F1 score of 0.723. The weakest class is blaming, with an F1 score of 0.308, partly because it has only eight examples in the test set.

**Table 3:** Model performance overview

| Model                        | Classes | Accuracy | Macro F1 | Test loss |
|------------------------------|---------|----------|----------|-----------|
| Five-label GBERT             | 5       | 0.579    | 0.499    | 1.155     |
| Merged-label GBERT           | 3       | 0.609    | 0.604    | 0.847     |
| Translated French evaluation | 3       | 0.649    | 0.648    | 0.797     |

The main error pattern is substantive rather than merely technical. The model often confuses shifting responsibility with explaining or sharing responsibility. This is theoretically plausible: both labels involve responsibility attribution, and the difference between distributing responsibility and shifting responsibility can be subtle in short media-conference segments.

This baseline result is informative even though it is not the preferred final model. It shows that the classifier can identify some categories clearly, especially credit claiming, but that the original label scheme asks the model to separate distinctions that may depend on pragmatic context. The low performance for blaming and praising should also be interpreted in light of their small sample sizes. With few examples, the classifier has limited opportunity to learn stable linguistic markers for these classes.

### Five-label GBERT model: confusion matrix

Rows are actual labels; columns are model predictions

| Actual label                      | Predicted label |                 |                                   |          |                         |
|-----------------------------------|-----------------|-----------------|-----------------------------------|----------|-------------------------|
|                                   | blaming         | credit claiming | explaining/sharing responsibility | praising | shifting responsibility |
| blaming                           | 2               | 2               | 3                                 | 0        | 1                       |
| credit claiming                   | 0               | 30              | 6                                 | 2        | 1                       |
| explaining/sharing responsibility | 1               | 4               | 29                                | 2        | 4                       |
| praising                          | 2               | 2               | 5                                 | 5        | 1                       |
| shifting responsibility           | 0               | 6               | 14                                | 0        | 11                      |

Figure 4. Confusion matrix for the five-label German GBERT model.

## 4.2 Merged-label model

The merged-label model improves both accuracy and macro F1. Accuracy increases from 0.579 to 0.609, while macro F1 increases from 0.499 to 0.604. The improvement in macro F1 is particularly important because it means that the broader label scheme improves performance across classes rather than only improving the majority class. The merged model also achieves a lower test loss of 0.847, compared with 1.155 for the five-label model.

### Merged-label GBERT model: model performance

Best checkpoint evaluated on held-out German test set

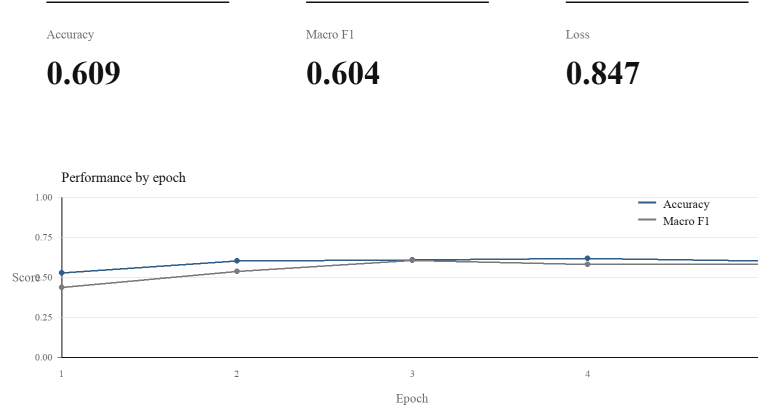


Figure 5. Performance summary for the merged-label GBERT model.

The confusion matrix shows that credit claiming remains comparatively identifiable. The model correctly classifies 25 of 39 credit claiming test segments. For implicit blame avoidance, it correctly classifies 36 of 55 segments. Explicit blame shifting is the most difficult merged category, with 20 correct classifications out of 39. Many explicit blame-shifting segments are predicted as implicit blame avoidance, suggesting that the boundary between explicit and implicit responsibility management remains challenging even after merging.

However, the merged model is not only numerically stronger, but also easier to interpret. Errors now mostly concern the boundary between implicit and explicit forms of blame management rather than five separate rhetorical subtypes. This three-class prediction can be used as a first-stage map of government communication and may be expanded by qualitative analyses.

### Merged-label GBERT model: confusion matrix

Rows are actual labels; columns are model predictions

| Actual label             | Predicted label |                         |                          |
|--------------------------|-----------------|-------------------------|--------------------------|
|                          | credit claiming | explicit blame shifting | implicit blame avoidance |
| credit claiming          | 25              | 6                       | 8                        |
| explicit blame shifting  | 8               | 20                      | 11                       |
| implicit blame avoidance | 4               | 15                      | 36                       |

Figure 6. Confusion matrix for the merged-label German GBERT model.

Table 4: Class-level performance of the merged-label German model

| Label                    | Support | Precision | Recall | F1    |
|--------------------------|---------|-----------|--------|-------|
| credit claiming          | 39      | 0.676     | 0.641  | 0.658 |
| explicit blame shifting  | 39      | 0.488     | 0.513  | 0.500 |
| implicit blame avoidance | 55      | 0.655     | 0.655  | 0.655 |

### 4.3 Testing translated French segments

The translated-French evaluation applies the merged-label German classifier to 222 originally French Swiss segments after machine translation into German. The result is encouraging: accuracy is 0.649 and macro F1 is 0.648. These values are slightly higher than the performance on the German test set. This should not be interpreted as evidence that the translated French segments are intrinsically easier. Translation can simplify or standardize wording, and the translated evaluation uses all translated French rows rather than only the original test split. Nevertheless, the result supports the feasibility of a translation-based cross-lingual pipeline.

### Translated French evaluation: model performance

Fine-tuned German model evaluated on translated French segments

Accuracy

**0.649**

Macro F1

**0.648**

Loss

**0.797**

Figure 7. Performance of the merged-label GBERT model on translated French segments.

The translated-French confusion matrix shows the same broad pattern as the German evaluation. Credit claiming is the easiest class, with 52 correct classifications out of 68. Explicit blame shifting is also reasonably stable, with 49 correct classifications out of 76. Implicit blame avoidance remains the most difficult class: 43 of 78 segments are classified correctly, while 19 are predicted as credit claiming and 16 as explicit blame shifting. This confirms that implicit blame avoidance is a middle category that can resemble either positive self-presentation or blame shifting depending on wording.

The translated-French result is notable because it does not show a collapse in performance. If the translation pipeline had distorted the categories severely, one would expect substantially lower accuracy and macro F1. Instead, the results are comparable to, and slightly higher than, the German test-set performance. This suggests that the major rhetorical distinctions survive translation well enough for model-assisted analysis. Still, the result should be validated with qualitative inspection of selected translated segments before being used for high-stakes substantive claims.

### Translated French evaluation: confusion matrix

Rows are actual labels; columns are model predictions

| Actual label             | Predicted label |                         |                          |
|--------------------------|-----------------|-------------------------|--------------------------|
|                          | credit claiming | explicit blame shifting | implicit blame avoidance |
| credit claiming          | 52              | 8                       | 8                        |
| explicit blame shifting  | 15              | 49                      | 12                       |
| implicit blame avoidance | 19              | 16                      | 43                       |

Figure 8. Confusion matrix for translated segments evaluated with the merged-

*label model.*

**Table 5:** Class-level performance on translated French segments

| Label                    | Support | Precision | Recall | F1    |
|--------------------------|---------|-----------|--------|-------|
| credit claiming          | 68      | 0.605     | 0.765  | 0.675 |
| explicit blame shifting  | 76      | 0.671     | 0.645  | 0.658 |
| implicit blame avoidance | 78      | 0.683     | 0.551  | 0.610 |

## 5 Discussion

The results have three main implications. First, transformer fine-tuning is feasible even with a relatively small manually annotated corpus, but the granularity of the label scheme matters. The five-label model learned meaningful patterns, but it struggled where the theoretical categories were close and the class sizes were small. Second, merging labels was not merely a technical fix for imbalance. It also aligned the classification task more closely with the substantive contrast between credit claiming, implicit blame avoidance, and explicit blame shifting. Third, the translation experiment suggests that the trained German model can be used as part of a multilingual workflow for Swiss data, provided that the resulting predictions are interpreted as classifications of translated text.

At the same time, several limitations remain. The corpus is small for transformer fine-tuning, especially compared with typical pretraining and large-scale supervised learning settings. Some categories have limited test support, and the existing train/test split should therefore be understood as an initial evaluation rather than a definitive performance estimate. A more robust future evaluation would use repeated cross-validation or a separately held-out set of media conferences. In addition, translation may alter rhetorical nuance. A French sentence that avoids blame implicitly may become more explicit, more neutral, or more standardized in German translation. The translated-French results are therefore promising but should not replace validation on manually coded French text if French-language inference becomes a central goal.

Another limitation concerns the target column. Targets such as citizens, sub-national level, and experts were retained in the dataset but not modeled as the primary prediction task. The target information is analytically valuable because blame avoidance is often about who is made responsible. A future model could therefore treat the task as multi-output classification, predicting both the strategy and the target. Alternatively, one could first classify credit claiming versus blame avoidance and then classify the target only for blame-avoidance segments.

A final interpretive point concerns the relationship between model performance and theory. The merged-label model performs better partly because it is statis-

tically easier, but that does not make the merge merely opportunistic. Label schemes should be both theoretically meaningful and empirically learnable. When a model repeatedly confuses two categories, this may indicate insufficient data, but it may also reveal that the categories overlap in actual language use. In this project, the improved merged model supports the view that implicit and explicit blame avoidance are more stable computational categories than the original five fine-grained labels.

## 6 Conclusion and Outlook

This project fine-tuned a German BERT-base model to classify governments’ credit claiming and blame avoidance strategies in COVID-19 media-conferences. The five-label model established a baseline but revealed systematic confusion between closely related categories. The merged-label model improved performance and produced a more interpretable three-class framework: credit claiming, implicit blame avoidance, and explicit blame shifting. The translated-French evaluation further showed that the merged German model can classify machine-translated French segments with reasonable accuracy.

For future work, the most promising next step is to enlarge the annotated dataset and test whether the merged-label model remains stable across additional time periods and countries. The model could then support large-scale analyses of how governments vary their blame and credit strategies over the course of a crisis. It would also be valuable to compare the translation pipeline with multilingual transformer models and with a French-specific fine-tuning strategy. Finally, qualitative inspection of the most common misclassifications should feed back into the coding scheme, because model errors often reveal precisely where theoretical categories overlap in real political communication.

The immediate practical output of the project is a reproducible pipeline: raw annotated Excel files are converted into clean CSV datasets, German segments are used for model fine-tuning, French segments can be translated into German, and all evaluations produce prediction files, metrics, and figures. This makes the project extensible. New annotated segments can be added to the raw data, the preprocessing and training scripts can be rerun, and changes in model performance can be tracked systematically over time.



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